

GridPulse — Milestone 3 Written Analysis

Team 9 · Intro to Cloud Computing · Containerize & Ship

1. Final image size and how we reduced it

Our backend image is **192 MB compressed** in Artifact Registry (804 MB on disk), down from **821 MB** with the Milestone 1 Dockerfile: **77% smaller**. The frontend image is 21.2 MB compressed (74.5 MB on disk). Three changes did it: (1) installing `xgboost-cpu` instead of `xgboost`, which drops the CUDA kernels and the `nvidia-nccl-cu12` dependency we never use, the largest saving; (2) multi-stage builds, so `pip`'s build stage and the frontend's Node toolchain and `node_modules` never ship, and the frontend runs on `nginx:alpine` with only the compiled files; (3) a `python:3.11-slim` base, no `pip` cache, and deleting the test suites bundled inside `pandas`, `numpy`, `scipy` and `scikit-learn`.

2. What `.dockerignore` excludes and why

- `.env` files, keys, credential JSON: secrets never reach the builder, so no `COPY` line can bake them into a layer.
- `node_modules`, `venv`, `__pycache__`: macOS binaries that would overwrite the Linux packages built inside the image and break the build.
- `.git`: full history, including anything ever committed by mistake.
- `uploads/`: runtime data that belongs in a Docker volume, not in the image.
- `frontend/dist`, `docs/`, `database/`, `backend/tests`, `*.tar.gz`: not needed at runtime; a stray repo tarball held a full copy of the project.

Result: a 1.2 MB build context containing only the files the Dockerfiles copy.

3. No secrets baked into the images

The only secret, `POSTGRES_PASSWORD`, lives in a `.env` file that is both `gitignored` and `dockerignored`; `docker-compose` injects `DATABASE_URL` at run time. Our script `deploy/verify_no_secrets.sh` confirmed on both images: no secret variables in the image config, the password absent from every `docker history` entry, no `.env` files anywhere in the filesystem, and 0 matches in a byte scan of every layer exported with `docker save`. Tested against a deliberately leaky image, the same script flagged all four problems.

4. Cost control

A \$50 monthly budget scoped to `gridpulse-509202` alerts at 10, 20, 25, 50 and 100%, counting usage before free-trial credits; October spend so far is \$2.42, covered by credits. The registry and VM share `us-central1` (no egress charges), and a cleanup policy deletes untagged images after 7 days.